

$$n=75$$

$$Avg = 10.7 \text{ Med. an} = 11$$

Quiz 3 October 2nd 2025 Chem 1127

Name (Full First then Last or -1 pts):

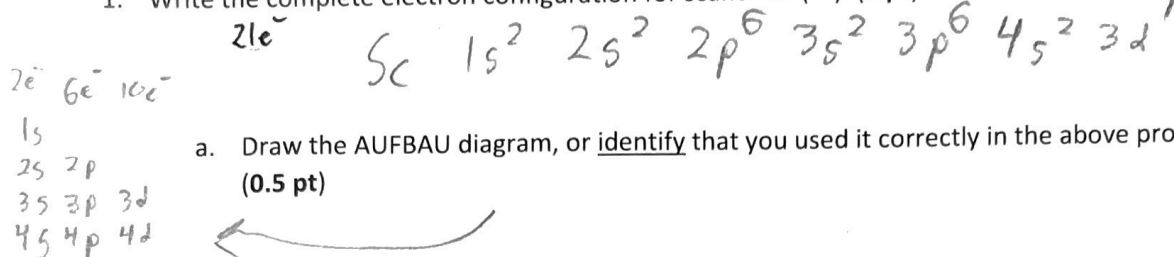
Key

Note: the quiz double sided

show your work

Points: ____/16₁₇

1. Write the complete electron configuration for scandium (Sc). (1 pt)

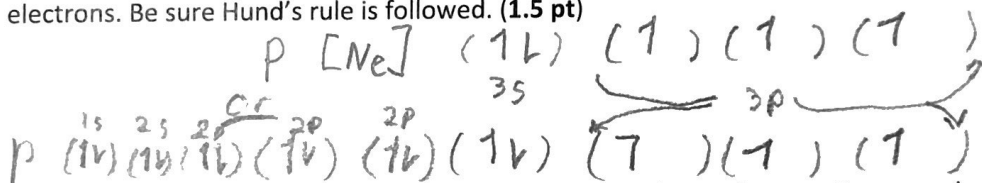


- a. Draw the AUFBAU diagram, or identify that you used it correctly in the above problem. (0.5 pt)

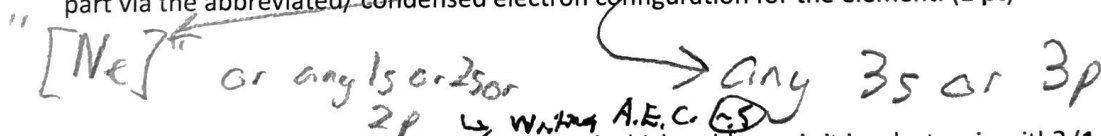
2. Write the abbreviated electron configuration (noble gas configuration) for strontium (Sr). (1 pt)



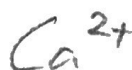
3. Write the orbital diagram for phosphorus (P), clearly label each orbital or sublevel (subshell), using boxes or a pair of parentheses to indicate each orbital. If you want, you may show only the valence electrons. Be sure Hund's rule is followed. (1.5 pt)



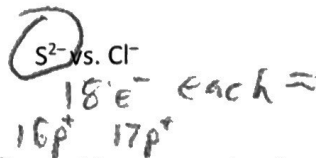
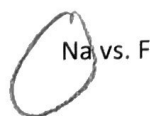
- a. On your orbital diagram, identify a core and a valence electron. You may also answer this part via the abbreviated/condensed electron configuration for the element. (1 pt)



4. What is the ionic charge when Ca becomes an ion, and which noble gas is it isoelectronic with? (1 pt)



5. In each of the pairs below, identify the larger species. (1 pt)



6. Electron affinity (EA) involves an electron being removed from a species, forming an anion. Identify what is incorrect with this definition of EA. (1 pt)



7. Identify which element is **more electronegative** in each pair. (1 pt)

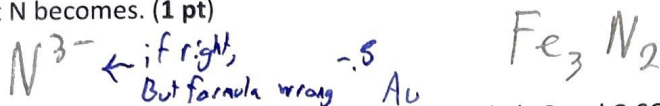


8. List 4 of the names & formulas with correct ionic charges of the 10 polyatomic ions that you were asked to memorize. (1 pt)

Slide 9

No passes - -1
complete

9. A3.151 Predict the chemical formula for the ionic compound formed by Fe^{2+} and the monatomic ion that N becomes. (1 pt)



- a. Provide the IUPAC names for FeCl_3 and Rb_2O and CrSO_4 . (1.5 pt)



Gold (III) chloride

Rubidium Oxide

Chromium (II) Sulfate

All or Nothing for each

10. Identify the following elements as metals (M) or nonmetals (NM). (1 pt)

He

Cl

Ti

H

NM

NM

M

NM

11. Which of the following compound(s) exhibits only covalent (molecular) bonding? Circle all correct choices. (1 pt)

a. Cl_2O_7

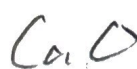
b. KCl

c. KNO_3

d. N_2O_4

→ only non metals & no ions

12. What are the correct formulas for calcium oxide and sodium chloride? Write them below. (1 pt)



13. Identify the two bonds below as ionic bonds (IB), polar covalent bonds (PCB), or nonpolar covalent bonds (NCB). (1 pt)

C - C

NCB

Fe - Br

IB

14. What is the ionic charge on Pb if it is in the ionic compound PbCl_4 ? (0.5 pt)

Pb^{4+}

Cl^-

Cl^-

Cl^-

Cl^-

Note that any given structure may not be drawn with the geometry that would result from VSEPR.

Quiz 4 Oct 9th 2025 Chem 1127

Name (Full First then Last or -1 pts):

Key

Note the quiz is front & back

SHOW YOUR WORK

Points: ____/16₁₇

1. Bonus. Describe what ancient civilizations used caltrops for, which have a tetrahedral geometry. Describe them or fill in the blank with the correct geometry. (0.5 pts)

spikes hurt & One always points "up".

2. Provide the correct chemical formula for the below name(s) or the correct IUPAC name (using the rules learned in class) for the given formula(s). (1.5 pt)

Cu^{2+} copper (II) chloride Cl^- dinitrogen trioxide SF_6 Sulfur Hexafluoride
 $CuCl_2$ N_2O_3

3. How many valence electrons do the elements below have? (1 pt)

Potassium

1

Oxygen

6

4. Draw the Lewis symbols for the elements below using dots to show the required electrons. (1 pt)

Carbon

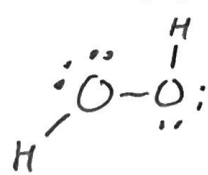
$\cdot \underset{\cdot}{\underset{\cdot}{\underset{\cdot}{C}}} \cdot$

Hydrogen

$H \cdot$

5. Draw the Lewis structure of H_2O_2 . Identifying the total # of valence electrons is a good move. (1.5 pt)

$$2 \times 1 + 6 \times 2 = 14$$



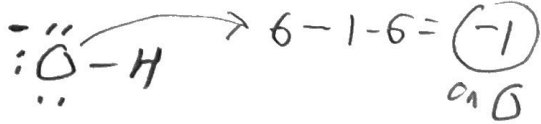
- a. The non-H atoms above follow the octet rule. Briefly define the octet rule, or circle "I used it correctly" below. (0.5 pt)

I used it correctly.

atoms want 8 e⁻ around them in Lewis structure

6. Draw the Lewis structure for OH^- . (1 pt)

$$6 + 1 - (-1) = 8$$



- a. Identify the formal charge for all atoms that have non-zero formal charges. (0.5 pt)

O has a negative 1 F.C.

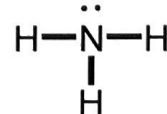
7. A Lewis structure of NH_3 is provided. Identify its electron-pair (electron domain) and molecular geometries. Note the geometries *may* be the same. (1 pt)

Electron-pair geometry

Molecular geometry

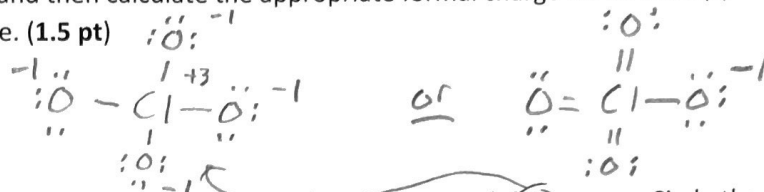
tetrahedral

trigonal pyramidal



Note that any given structure may not be drawn with the geometry that would result from VSEPR.

8. Draw the Lewis structure of ClO_4^- and then calculate the appropriate formal charge for all atom(s) that have a non-zero formal charge. (1.5 pt)



- a. Do all atoms in your structure above follow or are below the octet rule? **Yes** or **no**. Circle the correct bolded term. You must have a Lewis structure to be able to answer this question. (0.5 pt)

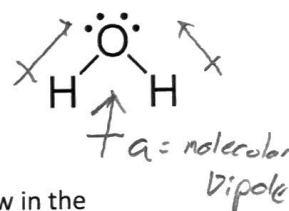
- b. Which of the elements in the above formula can be hypervalent (have an expanded octet)? Circle all that are correct below. (0.5 pt)

Cl O Neither can be hypervalent

9. The Lewis structure for H_2O is provided. Does this molecule possess polar covalent bond(s)? **Yes** or **No**. If Yes, clearly draw the bond dipoles using "plus arrows" like below. (1 pt)



Yes,



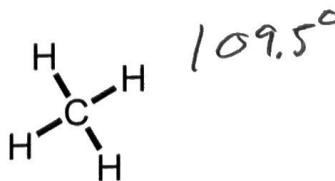
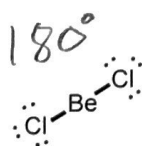
- a. Is this molecule **polar** or **nonpolar**? Circle the correct word. If it is polar, draw in the molecular dipole and label it "a". (1 pt)

- b. What is the strongest intermolecular force present in water? Circle it. (0.5 pt)
 Van der Waals Forces Dipole-Dipole Forces **Hydrogen-Bonding.**

- c. Is **HF** or **CH_3OCH_3** more soluble in water? Briefly explain your answer. (1 pt)

→ Like dissolves Like, both of these have Hydrogen Bonding

10. For the species below, identify the bond angles for each. (1 pt)



11. Identify how many atoms and lone pairs are present for a central atom (E) with the electron pair geometry provided below. (1 pt)

Atoms

5



Lone Pairs

1 lone pair ($2e^-$)